4.2 SAMPLED-DATA SYSTEMS

Now we move on to sampled-data systems, which are the most frequently encountered systems in practice. A sampled-data system is a system whose dynamics are described by a continuous-time differential equation, but the input only changes at discrete time instants, because (for example) the input is generated by a digital computer. In addition, we are interested in estimating the state only at discrete time instants. We are interested in obtaining the mean and covariance of the state only at discrete time instants. The continuous-time dynamics are described as

$$\dot{x} = Ax + Bu + w \tag{4.17}$$

From Chapter 1 we know that the solution of $x(t)$ at some arbitrary time, say t_k, is given as

$$x(t_k) = e^{A(t_k - t_{k-1})} x(t_{k-1}) + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)} [B(\tau)u(\tau) + w(\tau)]\, d\tau \tag{4.18}$$

Now assume that $u(t) = u_k$ for $t \in [t_k, t_{k+1}]$; that is, the control $u(t)$ is piecewise constant.[2] If we make the definitions

$$\begin{aligned} \Delta t &= t_k - t_{k-1} \\ x_k &= x(t_k) \\ u_k &= u(t_k) \end{aligned} \tag{4.19}$$

then Equation (4.18) becomes

$$x_k = e^{A\Delta t} x_{k-1} + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)} B(\tau)\, d\tau\, u_{k-1} + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)} w(\tau)\, d\tau \tag{4.20}$$

Now if we define F_k and G_k as

$$\begin{aligned} F_k &= e^{A\Delta t} \\ G_k &= \int_{t_k}^{t_{k+1}} e^{A(t_{k+1}-\tau)} B(\tau)\, d\tau \end{aligned} \tag{4.21}$$

then Equation (4.20) becomes

$$x_k = F_{k-1} x_{k-1} + G_{k-1} u_{k-1} + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)} w(\tau)\, d\tau \tag{4.22}$$

$e^{A(t_k-\tau)}$ is the state transition matrix of the system from time τ to time t_k. Now take the mean of the above equation, remembering that $w(t)$ is zero-mean, to obtain

$$\begin{aligned} \bar{x}_k &= E(x_k) \\ &= F_{k-1}\bar{x}_{k-1} + G_{k-1} u_{k-1} \end{aligned} \tag{4.23}$$

[2]This assumes that a first-order hold is used for the control inputs. Other types of holds can be used in sampled data systems, but in this book we assume that first-order holds are used.

We can use the previous equations to obtain the covariance of the state as

$$\begin{aligned}
P_k &= E[(x_k - \bar{x}_k)(x_k - \bar{x}_k)^T] \\
&= E\left[\left(F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}w(\tau)d\tau - \bar{x}_k\right)\left(\cdots\right)^T\right] \\
&= F_{k-1}P_{k-1}F_{k-1}^T + E\left[\left(\int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}w(\tau)\,d\tau\right)\left(\cdots\right)^T\right] \\
&= F_{k-1}P_{k-1}F_{k-1}^T + \int\int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}E\left[w(\tau)w^T(\alpha)\right]e^{A^T(t_k-\alpha)}\,d\tau\,d\alpha
\end{aligned} \tag{4.24}$$

Now, if we assume that $w(t)$ is continuous-time white noise with a covariance of $Q_c(t)$, we see that

$$E\left[w(\tau)w^T(\alpha)\right] = Q_c(\tau)\delta(\tau - \alpha) \tag{4.25}$$

This means that we can use the sifting property of the impulse function (see Problem 4.10) to write Equation (4.24) as

$$\begin{aligned}
P_k &= F_{k-1}P_{k-1}F_{k-1}^T + \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}Q_c(\tau)e^{A^T(t_k-\tau)}\,d\tau \\
&= F_{k-1}P_{k-1}F_{k-1}^T + Q_{k-1}
\end{aligned} \tag{4.26}$$

where Q_{k-1} is defined by the above equation; that is,

$$Q_{k-1} = \int_{t_{k-1}}^{t_k} e^{A(t_k-\tau)}Q_c(\tau)e^{A^T(t_k-\tau)}\,d\tau \tag{4.27}$$

In general, it is difficult to calculate Q_{k-1}, but for small values of $(t_k - t_{k-1})$ we obtain

$$\begin{aligned}
e^{A(t_k-\tau)} &\approx I \text{ for } \tau \in [t_{k-1}, t_k] \\
Q_{k-1} &\approx Q_c(t_k)\Delta t
\end{aligned} \tag{4.28}$$

■ EXAMPLE 4.2

Suppose we have a first-order, continuous-time dynamic system given by the equation

$$\begin{aligned}
\dot{x} &= fx + w \\
E[w(t)w(t+\tau)] &= q_c\delta(\tau)
\end{aligned} \tag{4.29}$$

First-order equations can be used to describe many simple physical processes. For example, this equation describes the behavior of the current through a series RL circuit that is driven by a random voltage $w(t)$, where $f = -R/L$. Suppose we are interested in obtaining the mean and covariance of the state $x(t)$ every Δt time units; that is, $t_k - t_{k-1} = \Delta t$. For this simple scalar example, we can explicitly calculate Q_{k-1} in Equation (4.27) as

$$\begin{aligned} Q_{k-1} &= \int_{t_{k-1}}^{t_k} \exp[f(t_k-\tau)] q_c \exp[f(t_k-\tau)]\, d\tau \\ &= \exp(2ft_k) q_c \int_{t_{k-1}}^{t_k} \exp(-2f\tau)\, d\tau \\ &= \exp(2ft_k) q_c \left[\frac{\exp(-2ft_{k-1}) - \exp(-2ft_k)}{2f}\right] \\ &= \frac{q_c}{2f}\left[\exp(2f(t_k - t_{k-1})) - 1\right] \\ &= \frac{q_c}{2f}\left[\exp(2f\Delta t) - 1\right] \end{aligned} \tag{4.30}$$

For small values of Δt, we can expand the above equation in a Taylor series around $\Delta t = 0$ to obtain

$$\begin{aligned} Q_{k-1} &= \frac{q_c}{2f}\left[\exp(2f\Delta t) - 1\right] \\ &= \frac{q_c}{2f}\left[\left(1 + 2f\Delta t + \frac{(2f\Delta t)^2}{2!} + \cdots\right) - 1\right] \\ &\approx \frac{q_c}{2f}\left[1 + 2f\Delta t - 1\right] \\ &= q_c \Delta t \end{aligned} \tag{4.31}$$

This matches Equation (4.28), which says that for small Δt we have $Q_{k-1} \approx q_c\Delta t$. The sampled mean of the state is computed from Equation (4.23) [noting that the control input in Equation (4.29) is zero] as

$$\begin{aligned} \bar{x}_k &= F_{k-1}\bar{x}_{k-1} + G_{k-1}u_{k-1} \\ &= \exp\left[f(t_k - t_{k-1})\right]\bar{x}_{k-1} + 0 \\ &= \exp(f\Delta t)\bar{x}_{k-1} \\ &= \exp(kf\Delta t)\bar{x}_0 \end{aligned} \tag{4.32}$$

We see that if $f > 0$ (i.e., the system is unstable) then the mean $\bar{x}_k$ will increase without bound (unless $\bar{x}_0 = 0$). However, if $f < 0$ (i.e., the system is stable) then the mean $\bar{x}_k$ will decay to zero regardless of the value of $\bar{x}_0$. The sampled covariance of the state is computed from Equation (4.26) as

$$\begin{aligned} P_k &= F_{k-1}P_{k-1}F_{k-1}^T + Q_{k-1} \\ &\approx (1 + 2f\Delta t)P_{k-1} + q_c\Delta t \\ P_k - P_{k-1} &= (2fP_{k-1} + q_c)\Delta t \end{aligned} \tag{4.33}$$

From the above equation, we can see that P_k reaches steady state (i.e., $P_k - P_{k-1} = 0$) when $P_{k-1} = -q_c/2f$, assuming that $f < 0$. On the other hand, if $f \geq 0$ then $P_k - P_{k-1}$ will always be greater than 0, which means that $\lim_{k\to\infty} P_k = \infty$.

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